Addis Ababa Science & Technology University

College of Natural & Applied Sciences

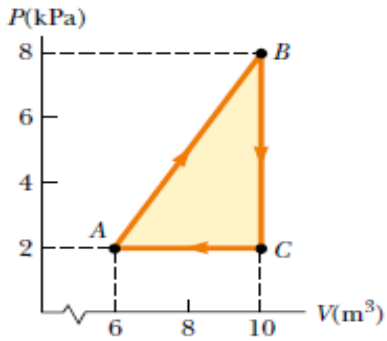
Department of Maths, Physics & Statistics

Worksheet 4 for General Physics (Phy1001)

1. Clearly distinguish among temperature, heat, and internal energy.
2. What is wrong with the following statement? "Given any two objects, the one with the higher temperature contains more heat."
1. The temperature of a silver bar rises by 10°C when it absorbs 1.23kJ of energy by heat. The mass of the bar is 525g. Determine the specific heat of silver.
2. Calculate the quantity of heat required to raise the temperature of 1 g of ice from -10^0C to 110^0C.
3. If 20 g steam initially at 100^0C is added to 60 g of ice initially at 0^0C, then find the final equilibrium temperature of the mixture.
4. The temperature of a silver bar rises by 10.0°C when it absorbs1.23 kJ of energy by heat. The mass of the bar is525 g. Determine the specific heat of silver.
5. A 50.0-g sample of copper is at 25.0°C. If 1 200 J of energy is added to it by heat, what is the final temperature of the copper?
6. A system absorbs 35J of heat and in the process it does 11J of work. (a) If the initial internal energy is 205J what is the final internal energy? (b) The system follows a different thermodynamic path to the same final state and does 15J of work, what is the heat transferred?
7. A gas is compressed at a constant pressure of 0.800atm from 9.00 L to 2.00 L. In the process, 400 J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?
8. A gas is confined to a vertical cylinder by a piston of mass 2 kg and radius 1 cm. When 5J of heat are added, the piston rises by 2.4 cm. Find: (a) the work done by the gas; (b) the change in its internal energy. Atmospheric pressure is 10^5Pa.
9. A thermodynamic system undergoes a process in which its internal energy decreases by 500 J. At the same time, 220 J of work is done on the system. Find the energy transferred to or from it by heat.
10. A gas is compressed at a constant pressure of 0.8atm from 9.00L to 2.0 L. In the process, 400J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?

11. A gas is taken through the cyclic process described in Figure below.

(a) Find the net energy transferred to the system by heat during one complete cycle.
(b) What If? If the cycle is reversed—that is, the process follows the path ACBA—what is the net energy input per cycle by heat?



12. How much heat energy is needed to convert a **1.5kg** of ice at **- 20°C** into steam at 100 **°C**. (Use c_{ice} =2090 J/Kg°C, L_{fusion}=3.33x10^5 J/Kg, c_{water} =4.19 x10^3 J/Kg°C, L_{vapor}=2.26x10^6 J/Kg, c_{steam} =2.01 x10^3 J/Kg°C)

13. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L
a) How much work is done by the gas during the expansion?

b) How much energy transfer by heat occurs with the surroundings in this process?

c) If the gas is returned to the original volume by means of an isobaric process, how much work

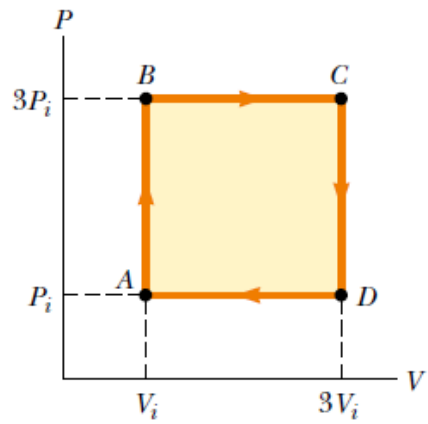
is done on the gas?

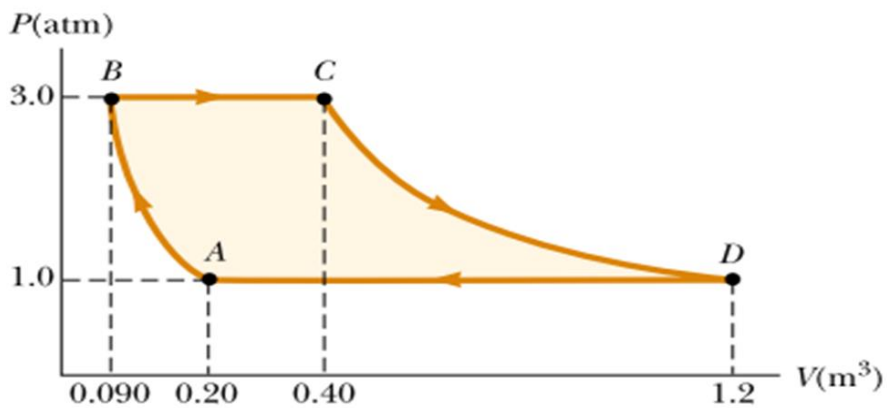
14. An ideal gas initially at $\mathbf{P_i}$ **,** $\mathbf{V_i}$, and $\mathbf{T_i}$ is taken through a cycle as shown in Figure below.

(a) Find the net work done on the gas per cycle.

(b) What is the net energy added by heat to the system per cycle?

(c) Obtain the numerical value of the work done per cycle for 1 mol of a gas initially at 0°c



15. A sample of an ideal gas goes through the process shown in the Fig below. From A to B, the process is adiabatic; from B to C, it is isobaric with 100 kJ of energy entering the system by heat. From C to D, the process is isothermal; from D to A, it is isobaric with 150 kJ of leaving the system by heat. Determine the difference in internal energy

$$E_{int,B} - E_{int,\ A}$$